

Jupyter Lab Plugin Installation

Jupyter Lab Plugin Installation

1. Install Node.js
2. Enable Plugins
3. Install Extensions

If you are using the factory image we provided, manual installation and configuration are not required.

Installing plugins can enhance Jupyter Lab's functionality and provide more tools and features.

1. Install Node.js

- Check Raspberry Pi architecture:

```
uname -a
```

```
yahboom@raspberrypi:~$ uname -a
Linux raspberrypi 6.1.0-rpi7-rpi-2712 #1 SMP PREEMPT Debian 1:6.1.63-1+rpt1 (2023-11-24) aarch64 GNU/Linux
```

Select the Node.js version suitable for ARMv8 architecture to install.

- Download Node.js:

```
https://nodejs.org/en/download/
```

Downloads

Latest LTS Version: **20.10.0** (includes npm 10.2.3)

Download the Node.js source code or a pre-built installer for your platform, and start developing today.

LTS Recommended For Most Users	Current Latest Features
 Windows Installer node-v20.10.0-x64.msi	 macOS Installer node-v20.10.0.pkg
 Source Code node-v20.10.0.tar.gz	

Windows Installer (.msi)

Windows Binary (.zip)

macOS Installer (.pkg)

macOS Binary (.tar.gz)

Linux Binaries (x64)

Linux Binaries (ARM)

Source Code

32-bit	64-bit	ARM64
32-bit	64-bit	ARM64
64-bit / ARM64		
64-bit	ARM64	
64-bit		
ARMv7	ARMv8	
node-v20.10.0.tar.gz		

Additional Platforms

Docker Image

Linux on Power LE Systems

Linux on System z

AIX on Power Systems

Official Node.js Docker Image	
	64-bit
	64-bit
	64-bit

- Extract:

After downloading the latest version of binary files from the official website, extract and rename the folder to node:

```
cd Downloads/
tar -xjf node-v20.10.0-linux-arm64.tar.xz
mv -i node-v20.10.0-linux-arm64 node
```

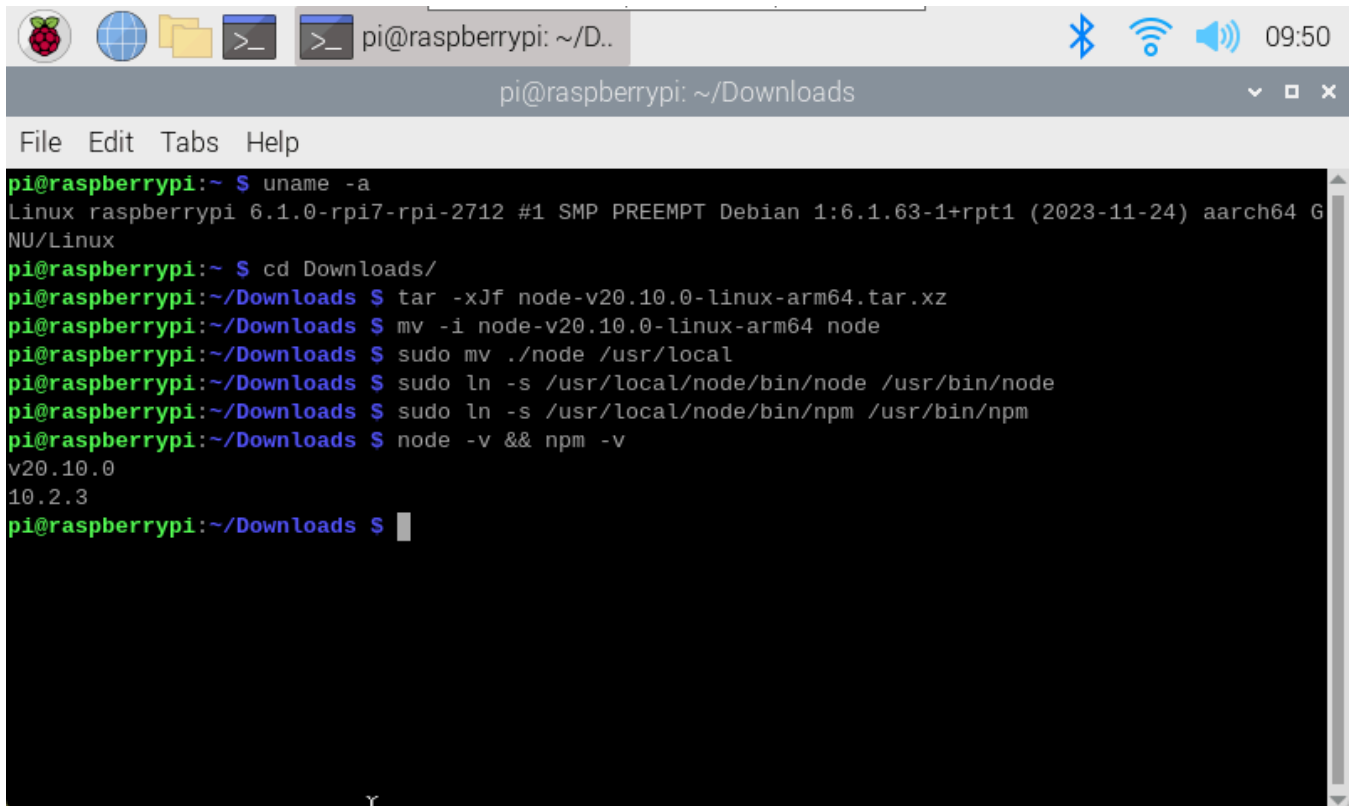
```
yahboom@raspberrypi:~ $ cd Downloads/
yahboom@raspberrypi:~/Downloads $ ls
node-v20.10.0-linux-arm64.tar.xz
yahboom@raspberrypi:~/Downloads $ tar -xjf node-v20.10.0-linux-arm64.tar.xz
yahboom@raspberrypi:~/Downloads $ mv -i node-v20.10.0-linux-arm64 node
yahboom@raspberrypi:~/Downloads $ ls
node node-v20.10.0-linux-arm64.tar.xz
```

- Move binary files and add symbolic links:

```
sudo mv ./node /usr/local
sudo ln -s /usr/local/node/bin/node /usr/bin/node
sudo ln -s /usr/local/node/bin/npm /usr/bin/npm
```

- Check if node and npm are installed successfully:

```
node -v && npm -v
```



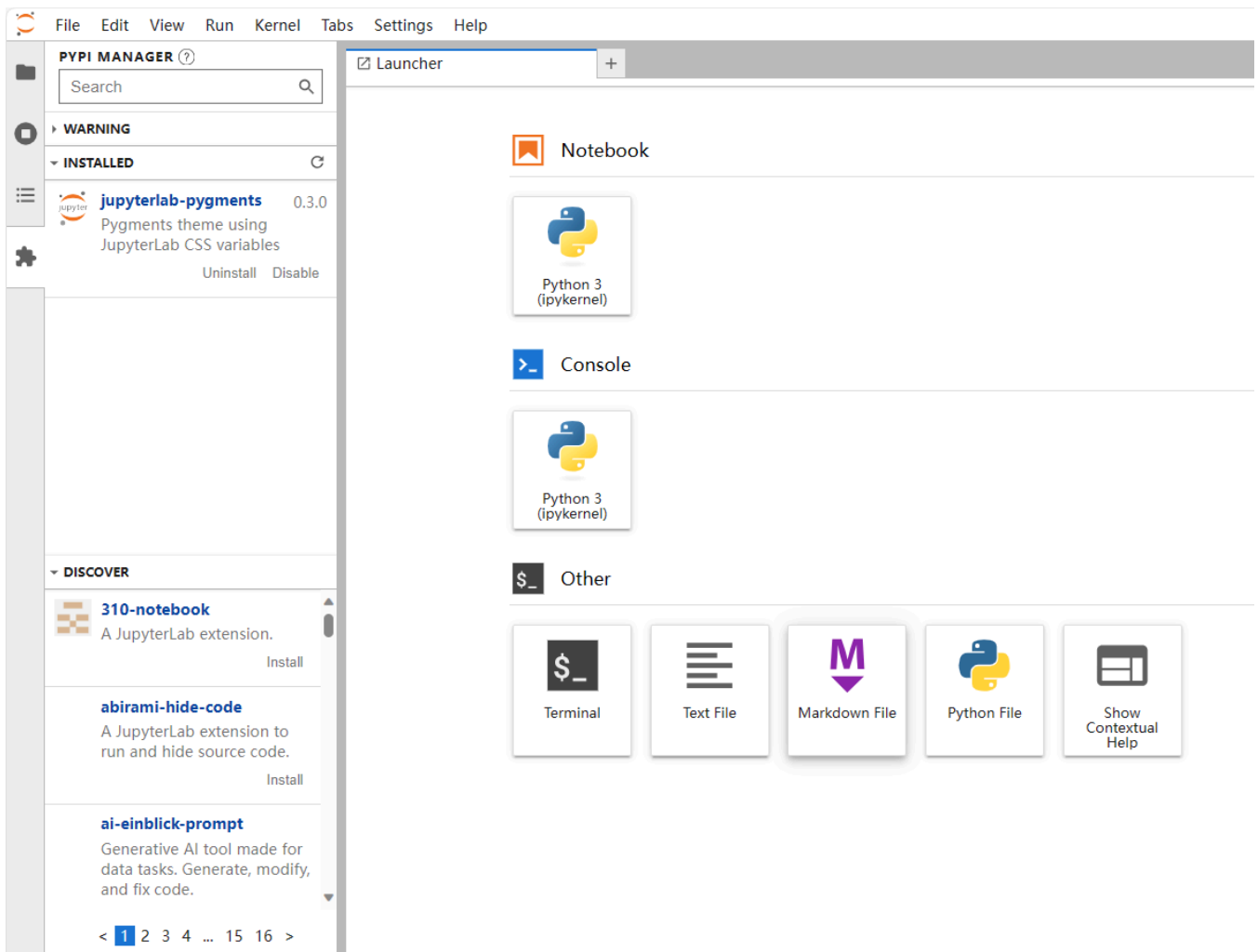
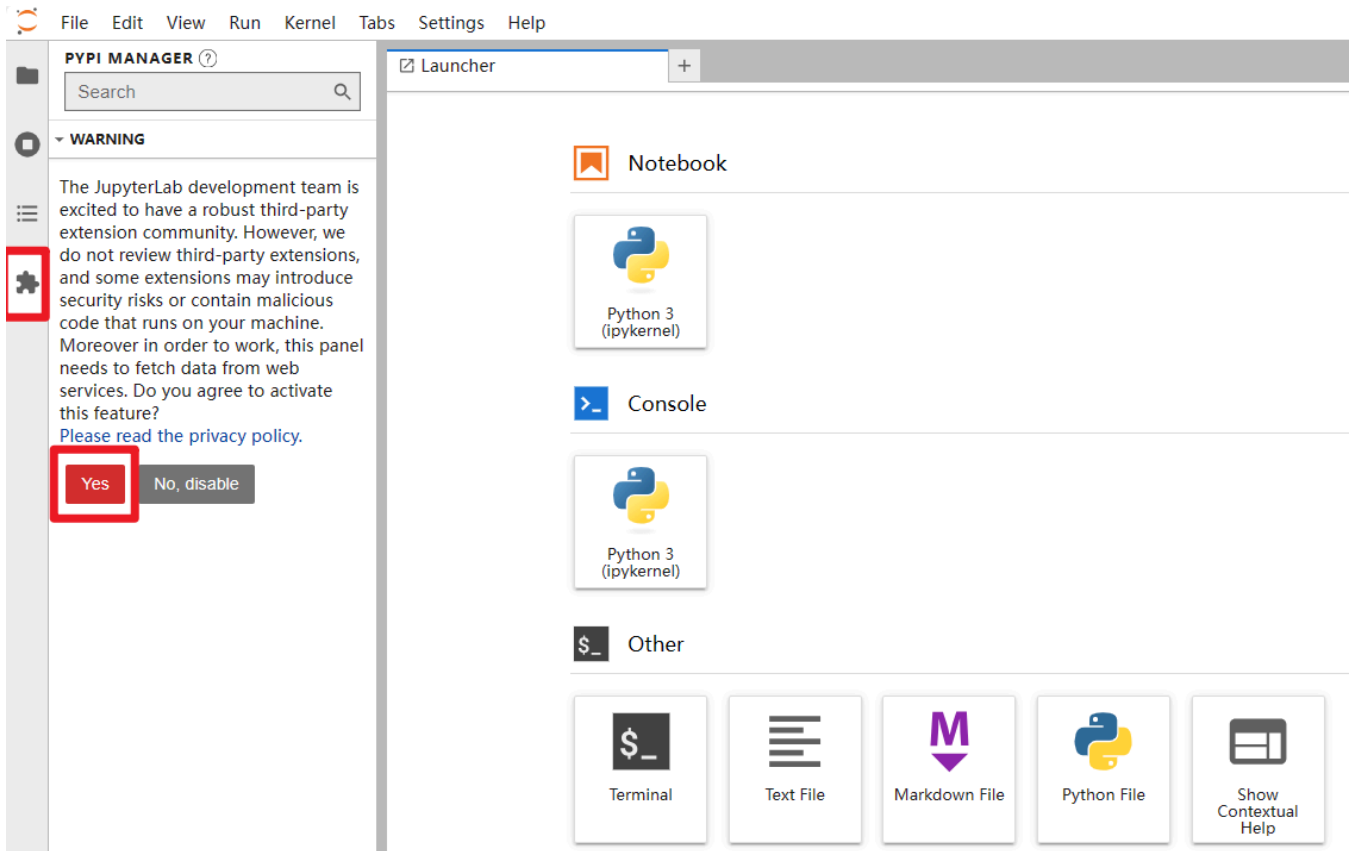
The screenshot shows a terminal window on a Raspberry Pi. The window title is "pi@raspberrypi: ~/Downloads". The terminal output shows the following commands and their results:

```
pi@raspberrypi:~ $ uname -a
Linux raspberrypi 6.1.0-rpi7-rpi-2712 #1 SMP PREEMPT Debian 1:6.1.63-1+rpt1 (2023-11-24) aarch64 GNU/Linux

pi@raspberrypi:~ $ cd Downloads/
pi@raspberrypi:~/Downloads $ tar -xJf node-v20.10.0-linux-arm64.tar.xz
pi@raspberrypi:~/Downloads $ mv -i node-v20.10.0-linux-arm64 node
pi@raspberrypi:~/Downloads $ sudo mv ./node /usr/local
pi@raspberrypi:~/Downloads $ sudo ln -s /usr/local/node/bin/node /usr/bin/node
pi@raspberrypi:~/Downloads $ sudo ln -s /usr/local/node/bin/npm /usr/bin/npm
pi@raspberrypi:~/Downloads $ node -v && npm -v
v20.10.0
10.2.3
pi@raspberrypi:~/Downloads $
```

2. Enable Plugins

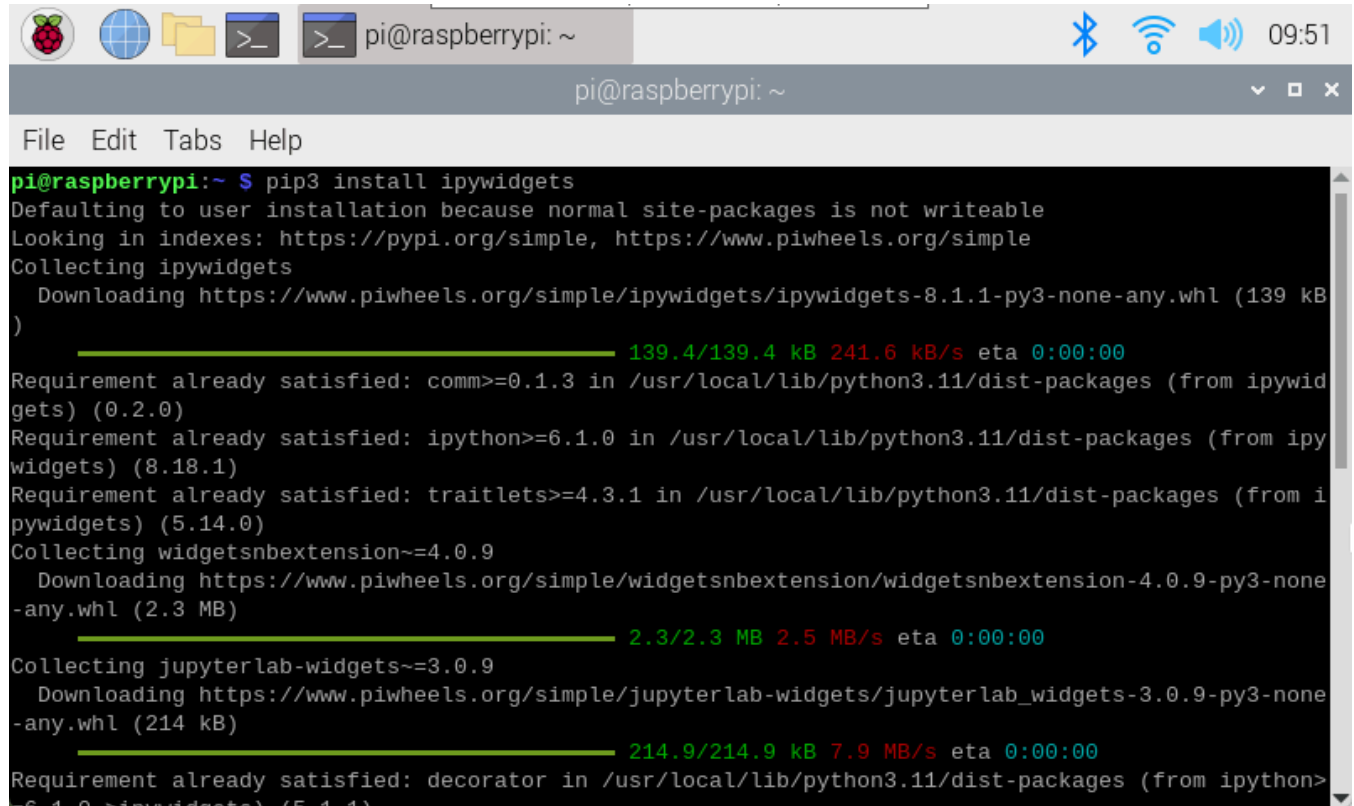
Enable third-party extension support in Jupyter Lab.



3. Install Extensions

```
pip3 install ipywidgets
```

Provides functionality for creating interactive widgets in Jupyter Lab.



The screenshot shows a terminal window titled 'pi@raspberrypi: ~' with a menu bar containing 'File', 'Edit', 'Tabs', and 'Help'. The terminal output shows the command 'pip3 install ipywidgets' being executed. The output indicates that the installation is defaulting to user installation because the normal site-packages is not writeable. It then shows the collection and download of ipywidgets (8.1.1) and its dependencies: comm (0.2.0), ipython (8.18.1), and traitlets (5.14.0). It also shows the collection and download of widgetsnbextension (4.0.9) and jupyterlab-widgets (3.0.9). The terminal output is as follows:

```
pi@raspberrypi:~ $ pip3 install ipywidgets
Defaulting to user installation because normal site-packages is not writeable
Looking in indexes: https://pypi.org/simple, https://www.piwheels.org/simple
Collecting ipywidgets
  Downloading https://www.piwheels.org/simple/ipywidgets/ipywidgets-8.1.1-py3-none-any.whl (139 kB)
    139.4/139.4 kB 241.6 kB/s eta 0:00:00
Requirement already satisfied: comm>=0.1.3 in /usr/local/lib/python3.11/dist-packages (from ipywidgets) (0.2.0)
Requirement already satisfied: ipython>=6.1.0 in /usr/local/lib/python3.11/dist-packages (from ipywidgets) (8.18.1)
Requirement already satisfied: traitlets>=4.3.1 in /usr/local/lib/python3.11/dist-packages (from ipywidgets) (5.14.0)
Collecting widgetsnbextension~=4.0.9
  Downloading https://www.piwheels.org/simple/widgetsnbextension/widgetsnbextension-4.0.9-py3-none-any.whl (2.3 MB)
    2.3/2.3 MB 2.5 MB/s eta 0:00:00
Collecting jupyterlab-widgets~=3.0.9
  Downloading https://www.piwheels.org/simple/jupyterlab-widgets/jupyterlab_widgets-3.0.9-py3-none-any.whl (214 kB)
    214.9/214.9 kB 7.9 MB/s eta 0:00:00
Requirement already satisfied: decorator in /usr/local/lib/python3.11/dist-packages (from ipython>=6.1.0->ipywidgets) (5.1.1)
```

```
pip3 install matplotlib
```

Provides functionality for creating various types of graphs and visualizations in Python, including line charts, scatter plots, histograms, etc.

```
pi@raspberrypi: ~  
File Edit Tabs Help  
pi@raspberrypi:~ $ pip3 install matplotlib  
Defaulting to user installation because normal site-packages is not writeable  
Looking in indexes: https://pypi.org/simple, https://www.piwheels.org/simple  
Collecting matplotlib  
  Downloading matplotlib-3.8.2-cp311-cp311-manylinux_2_17_aarch64.manylinux2014_aarch64.whl (11.4 MB)  
    _____ 11.4/11.4 MB 25.0 kB/s eta 0:00:00  
    _____ 11.4/11.4 MB 25.0 kB/s eta 0:00:00  
    _____ 11.4/11.4 MB 24.6 kB/s eta 0:00:00  
Collecting contourpy>=1.0.1  
  Downloading contourpy-1.2.0-cp311-cp311-manylinux_2_17_aarch64.manylinux2014_aarch64.whl (294 kB)  
    _____ 294.1/294.1 kB 21.3 kB/s eta 0:00:00  
Collecting cyclor>=0.10  
  Downloading https://www.piwheels.org/simple/cyclor/cyclor-0.12.1-py3-none-any.whl (8.3 kB)  
Collecting fonttools>=4.22.0  
  Downloading fonttools-4.47.0-cp311-cp311-manylinux_2_17_aarch64.manylinux2014_aarch64.whl (4.8 MB)  
    _____ 4.8/4.8 MB 20.3 kB/s eta 0:00:00  
Collecting kiwisolver>=1.3.1  
  Downloading kiwisolver-1.4.5-cp311-cp311-manylinux_2_17_aarch64.manylinux2014_aarch64.whl (1.4 MB)  
    _____ 1.4/1.4 MB 22.5 kB/s eta 0:00:00
```